SFU SCHOOL OF MECHATRONIC SYSTEMS ENGINEERING

MSE 220 Final Project

GE90 Turbine Blades: A Material Selection Study

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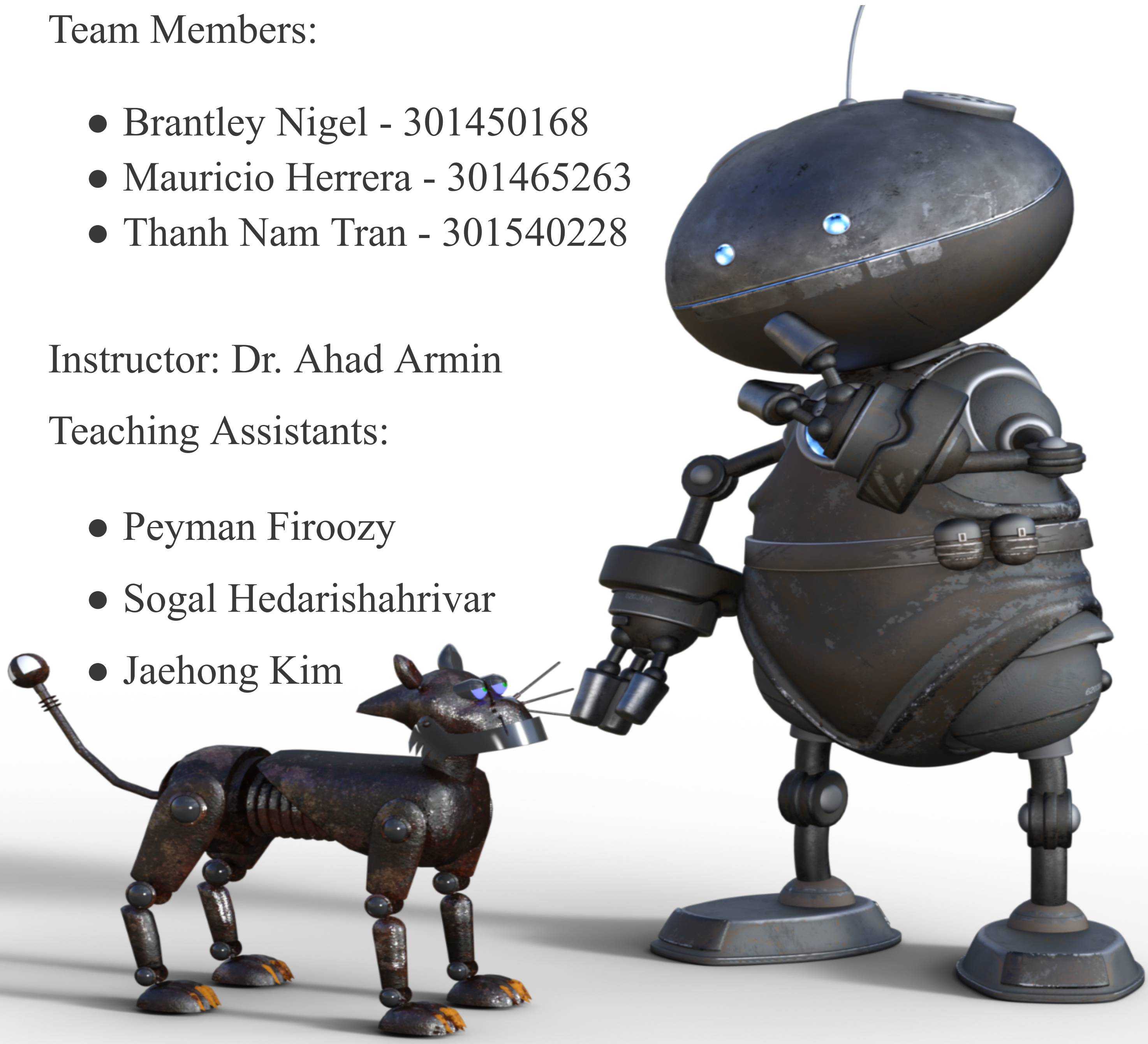

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Introduction

The primary objective of this project is to design a fan blade for an aircraft turbine engine, specifically for the General Electric GE90 - one of the marvels in modern aviation technology. This project not only aims at a practical application in aerospace engineering but also serves as an academic exercise in understanding the complexities and intricacies involved in material selection and engineering design.

The General Electric GE90 stands as a towering achievement in aviation history, known for being the largest physical engine ever used in the aviation industry. First introduced in the 1990s, the GE90 set new standards for power, efficiency, and reliability. It's a high-bypass turbofan engine that powers some of the largest aircrafts, including the Boeing 777. The significance of the GE90 lies not only in its size but also in its advanced technology, which has been pivotal in shaping contemporary aeronautics. It symbolizes a blend of engineering mastery and innovative design, pushing the boundaries of what was previously considered achievable in aircraft propulsion.

Central to the GE90, and indeed any jet engine, is the fan blade. These blades are critical components as they are the primary drivers of the engine's thrust. In a typical high-bypass turbofan engine like the GE90, a large fan at the front of the engine plays a crucial role. It works by drawing significant volumes of air into the engine, a portion of which is funneled into the core for combustion, while the majority is bypassed around the engine, producing most of the thrust. The design of these fan blades is a complex task that requires a meticulous balance between aerodynamic efficiency, material strength, and weight considerations. The fan blades must withstand extreme forces and temperatures, making their design and the materials used in their construction critical for the overall performance and safety of the engine. This project, therefore, focuses on designing a fan blade that meets these rigorous demands, combining theoretical knowledge with practical application to achieve an optimal solution.

Background and Project Requirements

The fan blade in a jet engine, particularly in the context of the General Electric GE90, plays a pivotal role in the engine's overall functionality and efficiency. The fan blade operates at the forefront of the engine, drawing in air with its rapid rotation, which is then bifurcated into two streams. One part enters the engine core for combustion, contributing to thrust and power, while the larger portion bypasses the core, providing the majority of the jet's thrust. This process highlights the importance of the fan blade's design and material integrity, as it directly impacts the engine's thrust capacity and, by extension, the aircraft's performance.

Design Specifications:

- Size and Shape: The specified dimensions for the fan blade are 900 mm in length and 300 mm in width, with a maximum thickness of 20 mm. The blade should possess an airfoil shape with a twist and taper along the radial axis. For weight reduction purposes, the blade may be designed with a hollow internal cavity. The blade's cross-section should be rectangular with the specified internal cavity.

Constraints and Challenges:

1. Tensile Stress Below Half of Ultimate Strength:
    a. Tensile Stress (σ) should be less than half of the Ultimate Strength (σ_{ts}).
    b. Tensile Stress (σ) = 0.11 × Density (ρ).
2. Fracture Toughness < $\sigma\sqrt{\pi a}$:
    a. a = Crack length (m).
3. Working Temperature Beyond 100°C.
4. Natural Frequency as High as Possible:
    a. ρ/E ratio should be as low as possible.
5. Cost + Fabrication Cost as Low as Possible:

   a. Fabrication cost = (external cavity surface area - internal cavity surface area) × 6000 = 0.16632.
   b. Cost = Price per kg/mass = Price per kg/(0.0033 × density).
6. Mass Calculation:
   a. Area of internal cavity (unused area) = Width × Length = 0.288 × 0.008 = 0.002304 (m^2).
   b. Total area = Width × Length = 0.3 × 0.02 = 0.006 (m^2).
   c. Area of external cavity (used area) = total area - area of internal cavity = 0.006 - 0.002304 = 0.003696 (m^2).
   d. Percentage of total area used = (Area of external cavity/total area) × 100 = 61.6%.
   e. Total volume = Width × Length × Thickness = 0.9 × 0.3 × 0.02 = 0.0054 (m^3).
   f. Volume used = total volume × 0.616 = 0.0054 × 0.616 = 0.0033264 (m^3).
   g. Mass = volume used × density.

In summary, the design and selection of the fan blade material for the GE90 engine must carefully balance these constraints and challenges. The material must be lightweight, cost-effective, resilient to tensile stress and fatigue, resistant to damage and vibrational loads, and capable of performing under high-temperature conditions. This intricate balance of requirements necessitates a thorough and methodical approach to material selection, ensuring the fan blade's optimal performance in the demanding environment of a jet engine.

Material Selection Analysis

Tensile Stress Below Half of Ultimate Strength

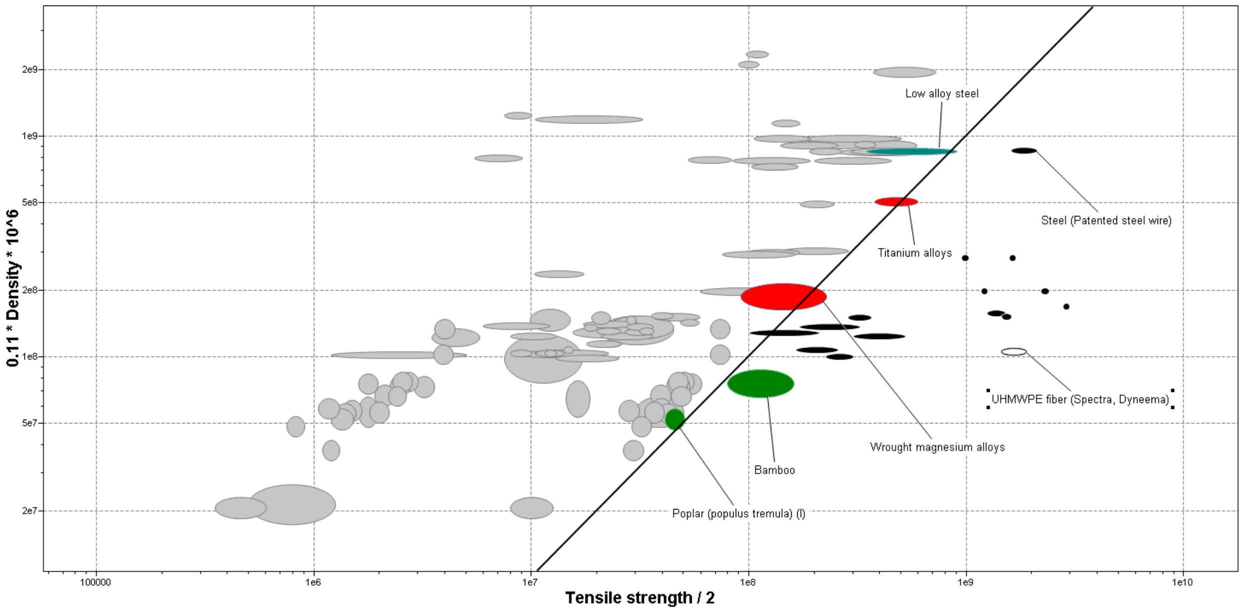

Figure 1 : Analysis of Tensile stress in Edupack.

Analysis

- The crucial factor in the material selection process for the fan blade is the tensile stress. This aspect is particularly significant as it directly relates to the material's ability to endure the stresses of 100,000 engine flights. The analysis in this phase focuses on how well materials can withstand tensile stress, specifically compared to half of their ultimate strength.
- The rationale for using half the ultimate strength as a benchmark is to ensure that the material operates well within its elastic region. This consideration is vital because a larger elastic region enhances the material's ductility, allowing it to absorb and withstand heavy cyclic loads more effectively. This property is crucial for aircraft components that undergo repeated stress cycles during their operational lifetime.
- In the analysis conducted using Granta EduPack, materials were evaluated based on their performance against the defined tensile stress criterion. The correlation line on the graph served as a benchmark to determine the materials' suitability. Materials intersecting or lying above this correlation line were deemed capable of withstanding tensile stresses at

levels less than half of their ultimate strength, thereby ensuring adequate durability and reliability under operational conditions.

Materials that Passed the Tensile Stress Criterion:

- The materials that successfully met the tensile stress criterion, as evidenced in the EduPack analysis, are as follows. These materials are not only capable of withstanding substantial tensile stresses but also align with the previously evaluated criteria of fracture toughness and temperature resistance:
    - Acrylic fiber (PAN)
    - Aramid fiber (Kevlar 49)
    - Bamboo
    - Carbon fiber (HM carbon)
    - Carbon fiber (HS carbon)
    - Cellulosics fiber (Rayon)
    - Glass fiber (C-glass)
    - Glass fiber (E-glass)
    - Low alloy steel
    - PBO fiber (Zylon)
    - PLA fiber (Ingeo)
    - Polyamide fiber (Nylon-6)
    - Polyarylate fiber (Vectran)
    - Polyester fiber (Dacron)
    - Polypropylene fiber
    - Poplar (populus tremula) (l)
    - Steel (Patented steel wire)
    - Titanium alloys
    - UHMWPE fiber (Spectra, Dyneema)

- Wrought magnesium alloys

- A comparative review was conducted against the previously analyzed graphs to ensure consistency and compatibility across different constraints. The result of this comprehensive analysis confirms that these materials meet all the physical requirements necessary for the fan blade's functionality and durability.

Fracture Toughness < $\sigma\sqrt{\pi a}$:

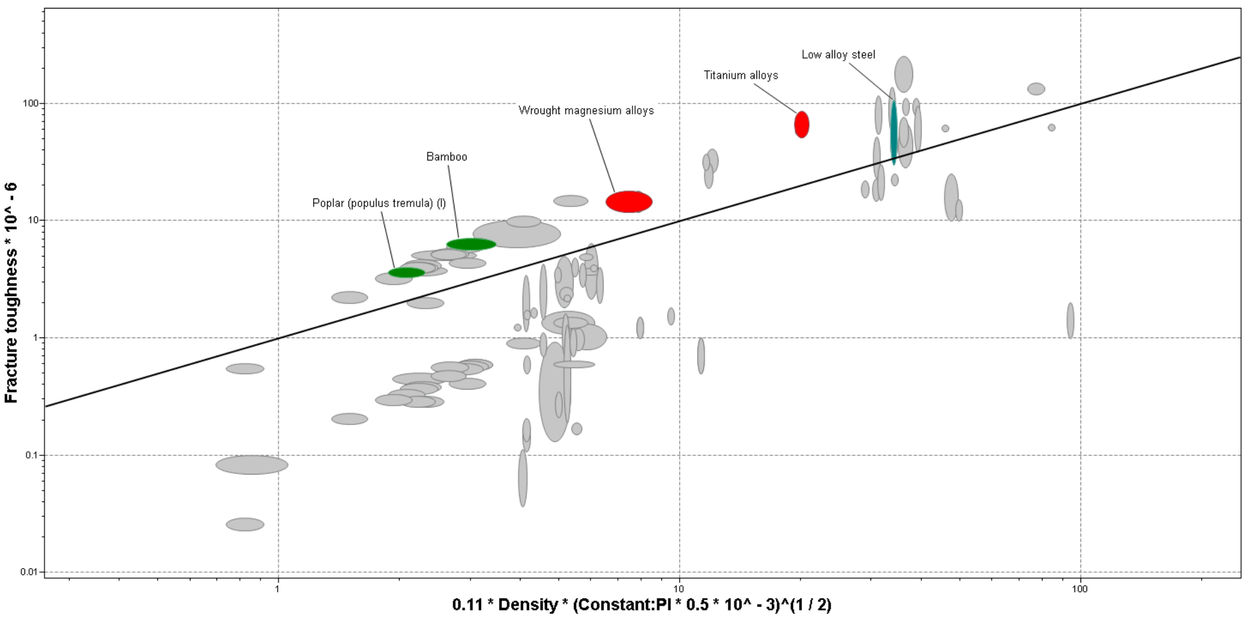

Figure 2 : Analysis of Fracture Toughness in Edupack.

Analysis

- The integrity of the turbine blade's exterior, especially at high altitudes, is paramount. It must resist fracturing under the immense stresses encountered during flight. Consequently, we need to consider fracture toughness. The aim is to ensure that the material chosen for the fan blade does not exhibit any fractures, even minor ones, under operational conditions.
- In the analysis, the axis was set to compare fracture toughness against density, factoring in the crack length. This setup in Granta EduPack allowed for a clear visualization of

materials that can withstand the specified crack length. The focus was on a 0.25mm internal crack length, a size likely to result from the centrifugal forces experienced by the blade during operation.
- The correlation line on the graph served as a critical indicator. Materials that remained above this line were deemed capable of withstanding the internal crack pressures without succumbing to fracture. This criterion was pivotal in shortlisting materials that could potentially be used for the fan blade, ensuring they meet the minimum standard for fracture toughness.

Materials that Passed the Fracture Toughness Criterion:

- The materials that remained above the correlation line and hence are potential candidates for the fan blade include:
    - Bamboo
    - Low alloy steel
    - Poplar (populus tremula) (l)
    - Titanium alloys
    - Wrought magnesium alloys

This analysis, grounded in the fracture toughness criterion, successfully narrowed down the range of suitable materials for the fan blade, ensuring the selected material can withstand the operational stresses without fracturing.

Working at Temperature Beyond 100°C

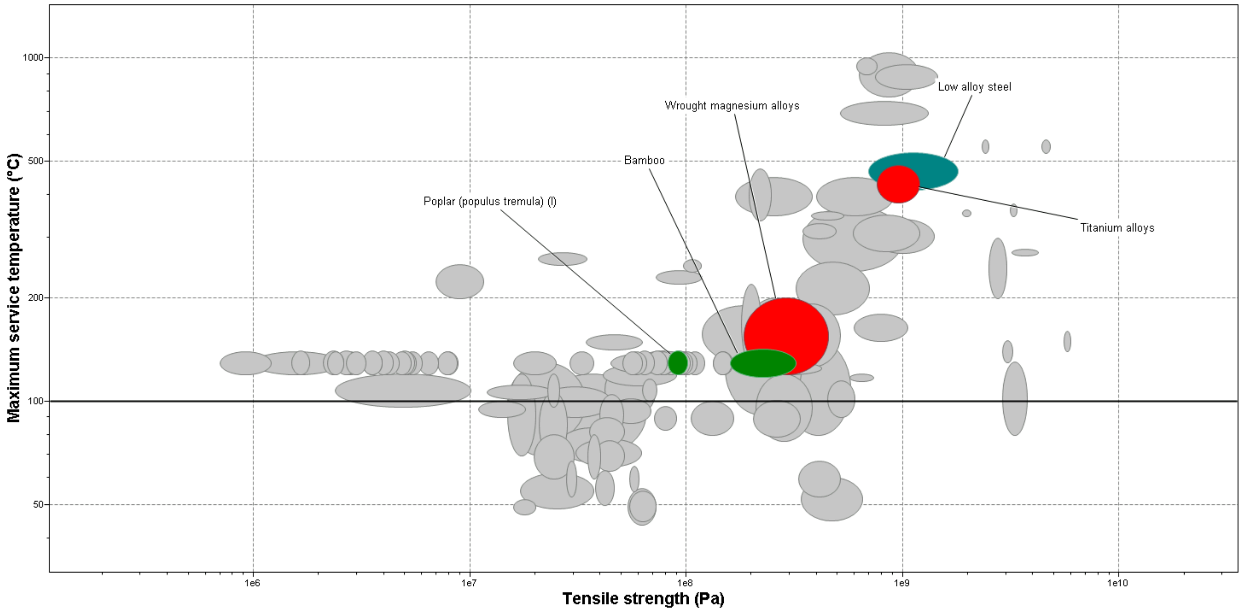

Figure 3: Analysis Of Temperature in Edupack.

Analysis

- Following the assessment of materials for fracture toughness, the next critical constraint to consider is heat resistance. Given the high-speed operation of aircraft in the air, the temperature buildup on the exterior components, particularly the fan blades, is a significant concern. Excessive temperatures can lead to the failure of critical functions, posing potential dangers.
- To address this, a minimum working temperature threshold was set in the analysis. Using Granta EduPack, a material chart was generated to assess which materials could effectively operate at temperatures beyond 100°C. This temperature criterion is particularly crucial because it directly impacts the material's ability to maintain its integrity and performance under the operational thermal conditions of an aircraft.
- The correlation line in the material chart plays a crucial role in this analysis. It demarcates the materials that can withstand the set minimum temperature from those that cannot. Materials that fall above this line are considered capable of maintaining their structural and functional integrity at temperatures exceeding 100°C.

Materials that Passed the Temperature Resistance Criterion:

- The materials that successfully met the temperature resistance criterion, as per the analysis in EduPack, are listed below. These materials have demonstrated their ability to function effectively at and beyond the minimum required temperature of 100°C:
    - Bamboo
    - Low alloy steel
    - Poplar (populus tremula) (l)
    - Titanium alloys
    - Wrought magnesium alloys

This analysis, focusing on temperature resistance, further refines the list of suitable materials for the fan blade, ensuring that the final selection can withstand the operational temperatures of an aircraft's engine without compromising safety or performance.

Price and Weight as Low as Possible

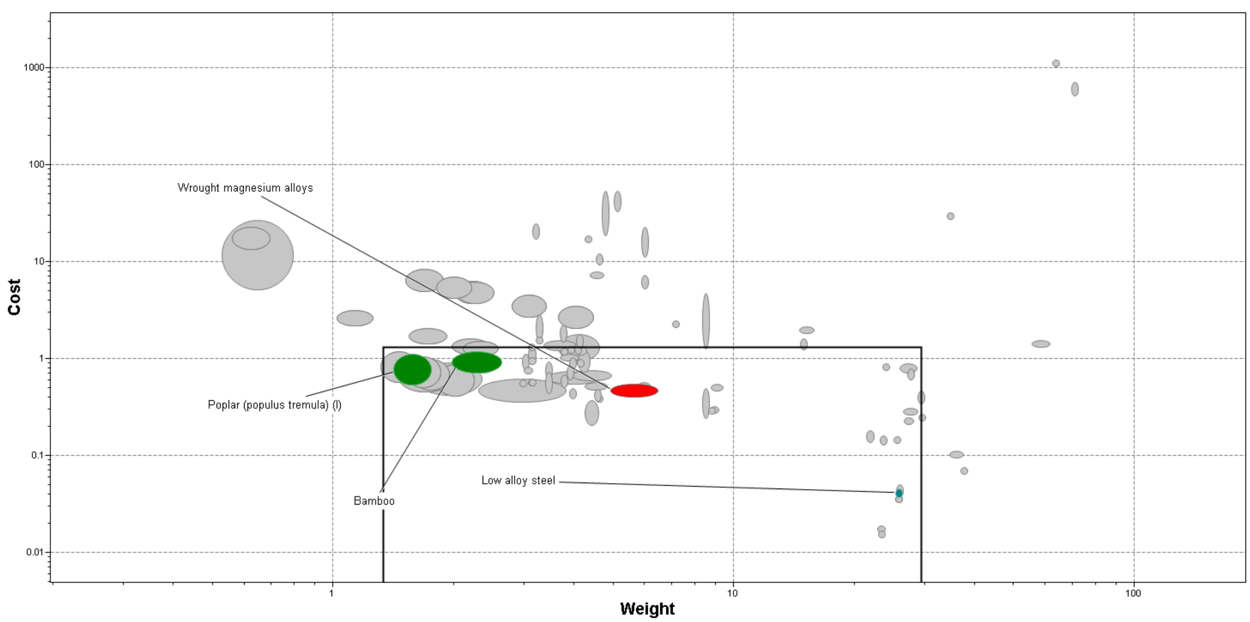

Figure 4: Analysis of Price in Edupack.

Analysis

- In addition to the mechanical and physical properties, the cost of materials is a crucial aspect to consider in the selection process, especially for commercial applications where budget constraints are significant. The objective is to identify materials that not only meet the technical requirements but are also economically viable.
- Utilizing Granta EduPack, an analysis was conducted to determine the most cost-effective materials among those that had already passed the previous constraints (fracture toughness, temperature resistance, tensile stress). The focus was on finding materials that offer the best balance between performance and price.
- The analysis revealed that bamboo, magnesium, low-alloy steel, and Poplar (typical along grain) are the most cost-effective options. These materials, while being economical, still comply with the necessary constraints for the fan blade's functionality and longevity.
- It's noteworthy that while materials like CFRP (Carbon Fiber Reinforced Polymer) and titanium alloys offer excellent performance, their higher costs could be prohibitive, particularly for large-scale commercial applications. In scenarios where budget restrictions are stringent, it's imperative to consider materials that provide a practical compromise between cost and performance.

Materials that Passed the Cost Criterion:

- The materials that successfully met the cost-effectiveness criterion are:
    - Bamboo
    - Low alloy steel
    - Poplar (populus tremula) (l)
    - Wrought magnesium alloys

- These materials were selected based on their lower price points while still maintaining compliance with the physical and mechanical requirements previously established. The selection reflects a strategic approach to material choice, one that balances technical specifications with economic feasibility.

Natural Frequency Considerations

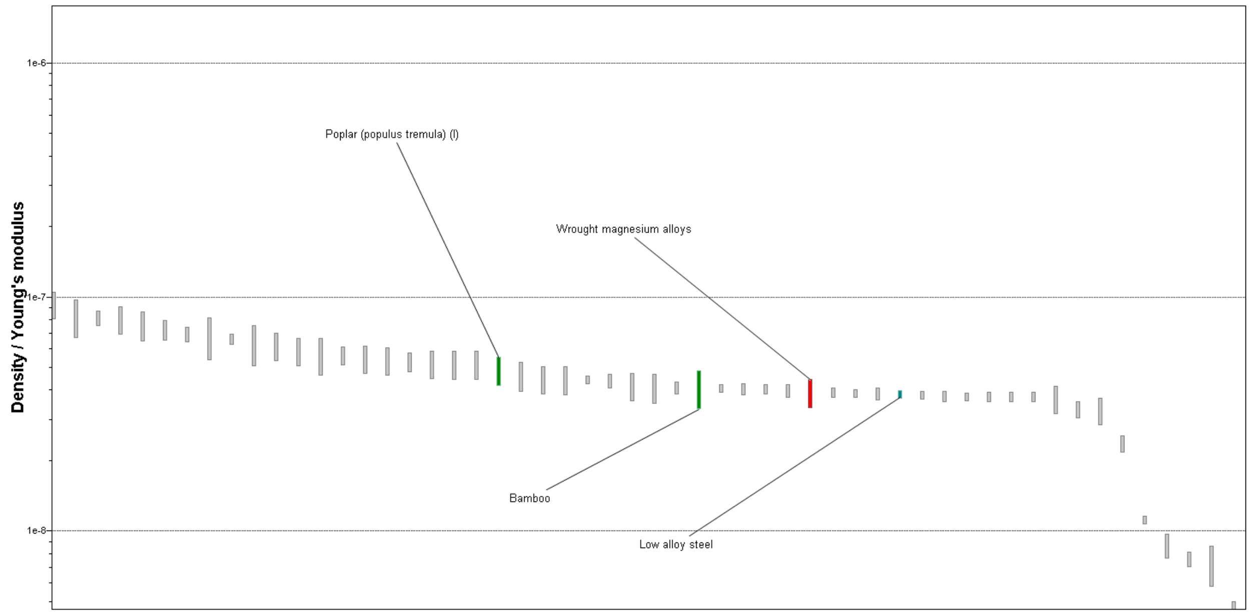

Figure 5: Analysis of Natural Frequency in Edupack.

Analysis: Considering the need for high natural frequency to minimize vibrational fatigue, these materials offer an optimal combination of stiffness and low density, suitable for the demanding conditions of aircraft turbine blades:

- Bamboo
- Low alloy steel
- Poplar (populus tremula) (l)
- Wrought magnesium alloys

Environment and Energy Consumption

Calculations

- Material Densities and Masses:
    - Poplar (Typically Along Grain): Density = 470 kg/m³, Mass = 1.56 kg
    - Bamboo: Density = 650 kg/m³, Mass = 2.16216 kg
    - Low Alloy Steel: Density = 7800 kg/m³, Mass = 25.96 kg
- Project Specifications:

- Number of Blades: 1000
- End of Life: Landfill
- Product Life: 2 Years
- Usage: 300 Days/Year
- Travelled Distance: 10,000 km/Day
- Sea Shipping Distance: 3.8*10^7 km

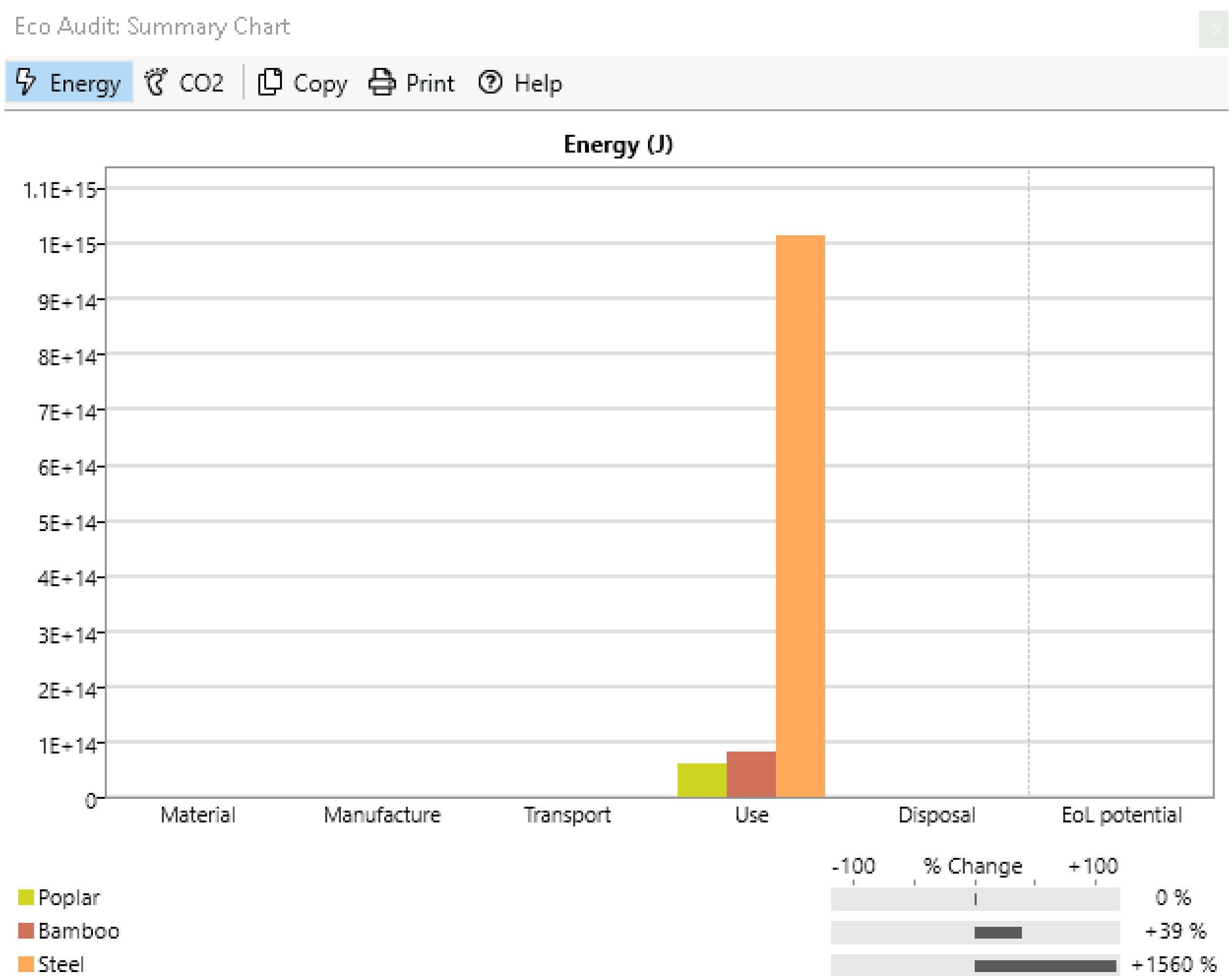

Figure 6: Analysis of Environmental and Energy Consumption in Edupack.

Analysis

- The environmental impact and energy consumption associated with the production and use of the fan blades is a critical factor in the material selection process. This aspect is particularly important in the context of sustainable engineering and eco-friendly practices.
- An analysis was conducted using Granta EduPack to compare the energy consumption for manufacturing 1000 blades from each of the shortlisted materials – bamboo, Poplar, and steel. The evaluation took into account the entire lifecycle of the blades, from production to end-of-life disposal, factoring in the product life, usage patterns, and the transportation distances involved.
- The analysis revealed that low alloy steel, while meeting the physical and mechanical requirements, has a significantly higher energy consumption compared to bamboo and Poplar. This increased energy requirement not only translates to higher costs but also results in a larger environmental footprint.
- Comparing bamboo and Poplar, the study showed that Poplar has a lower energy consumption than bamboo. This lower energy requirement makes Poplar a more environmentally sustainable option. The graph from EduPack provided a visual representation of these findings, clearly indicating that Poplar, among the evaluated materials, has the least environmental impact in terms of energy consumption.
- Considering the environmental factors, Poplar emerges as the most suitable material for the fan blades. It aligns well with the eco-criteria, offering a balance between mechanical functionality and environmental sustainability.

Final Material Analysis

As most of the results lean towards the usage of Bamboo, the environmental aspect chooses it towards Poplar. However, bamboo still offers the physical qualities that are needed the most on building the blades of the plane. Despite Poplar's environmental advantages, bamboo emerges as the more favorable material when considering the overall requirements of the project.

Bamboo's Superiority in Key Aspects:

- **Heat Resistance:** Bamboo exhibits greater heat resistance compared to Poplar. This property is crucial for the fan blade, ensuring better performance and maintenance under high-temperature conditions typically experienced during flights. The ability of bamboo to maintain its structural integrity at elevated temperatures makes it particularly suited for the demanding operational environment of aircraft engines.
- **Tensile Strength:** The tensile strength of bamboo is higher than that of Poplar. This characteristic is vital for the fan blade's ability to endure heavy cyclic loads, a common scenario in aviation engines. Bamboo's enhanced tensile strength ensures that the blade can withstand the rigorous stress cycles of frequent takeoffs, flights, and landings without succumbing to material fatigue.
- **Impact Absorption:** Bamboo's superior impact absorption capabilities make it more suitable for withstanding the various stresses, including the possibility of foreign object impacts, that a turbine blade would encounter. This resilience to physical impacts is critical for ensuring the long-term reliability and safety of the fan blades.
- **Cost-Effectiveness:** When considering the price per kilogram, bamboo is more economical than Poplar. This cost efficiency is a significant factor, especially for commercial applications where budget constraints are often a primary concern. Bamboo's affordability, combined with its mechanical properties, makes it a particularly attractive material choice for the mass production of turbine blades.

Conclusion

The research and analysis process has effectively narrowed the selection to bamboo. Through a balanced consideration of mechanical properties, environmental impact, and cost, bamboo stands out as the material that best meets the multifaceted demands of turbine blade production. Its combination of strength, durability, heat resistance, and cost-effectiveness aligns perfectly with the project's objectives, making it the optimal choice for this application. The selection of bamboo not only addresses the technical and economic requirements of the turbine blade but also demonstrates a commitment to selecting materials that offer a harmonious blend of performance and practicality.